NOETHERIAN APPROXIMATION OF A FINITELY PRESENTED FLAT SHEAF OVER A FINITELY PRESENTED MORPHISM OF AFFINES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 9789-9797 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 9799-9812, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Characterizations of finite presentation for algebras and modules, flatness descent to modules, and the Noetherian approximation lemma for flat modules over a limit ring (Algebra); cited inputs, with the three hypothesis verifications written out in the retained span.

Source reference. Original lemma 02JO: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9812–9812. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05FC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9767–9774. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01U4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9812–9812. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01TQ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9812–9812. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01PC: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 9812–9812. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to S$ be a morphism of affine schemes, which is of finite presentation. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module of finite presentation and flat over S. Then we may choose a diagram as in Lemma 05FC and sheaf $\mathcal{F}_0$ such that in addition $\mathcal{F}_0$ is flat over S_0.*

Proof. Write $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, and $\mathcal{F} = \widetilde{M}$. As f is of finite presentation we see that B is of finite presentation as an A-algebra, see Morphisms, Lemma 01TQ. As $\mathcal{F}$ is of finite presentation over $\mathcal{O}_X$ we see that M is of finite presentation as a B-module, see Properties, Lemma 01PC. As $\mathcal{F}$ is flat over S we see that M is flat over A, see Morphisms, Lemma 01U4. Thus the lemma follows from Algebra, Lemma 02JO. □